

LAPORAN UJIAN AKHIR SEMESTER (UAS) KECERDASAN BUATAN

DOSEN PENGAMPU:

Teuku Rizky Noviandy, S.Kom., M.Kom.



DI SUSUN OLEH:

Najwa Shalsabila (24146031)

PROGRAM STUDI (S1) SISTEM INFORMASI

FAKULTAS TEKNIK

UNIVERSITAS ABULYATAMA

ACEH BESAR

2026

1. PENDAHULUAN

Laporan ini berisi hasil implementasi dan pengujian algoritma pencarian/optimasi berbasis Algoritma Genetika (Genetic Algorithm) untuk menyelesaikan permasalahan penjadwalan mata kuliah. Algoritma genetika bekerja secara iteratif melalui tahapan inisialisasi populasi, perhitungan nilai fitness (fungsi evaluasi), seleksi orang tua, perkawinan silang (crossover), dan mutasi untuk menemukan solusi terbaik dengan jumlah konflik minimal.

2. DESKRIPSI TUGAS & KETENTUAN OUTPUT

Berdasarkan ketentuan tugas Ujian Akhir Semester (UAS), program yang dibangun wajib memenuhi kriteria keluaran (output) sebagai berikut:

- **Menampilkan jadwal terbaik yang diperoleh beserta alokasi Mata Kuliah (MK), Ruang, Waktu, dan Dosen.**
- **Menampilkan nilai Fitness Terbaik dan Jumlah Konflik yang tersisa pada hasil evaluasi akhir.**
- **Mencetak perkembangan nilai Max Fitness dan Avg Fitness pada setiap generasi (generasi 1 hingga n).**

3. HASIL EKSEKUSI PROGRAM & TAMPILAN OUTPUT

Berikut adalah bukti eksekusi program beserta screenshot hasil keluaran (output) yang telah dikembangkan:

3.1. Output Jadwal Terbaik (Poin 6)

Program berhasil menggenerasi alokasi jadwal terbaik untuk setiap Mata Kuliah (MK) dengan pembagian ruang, waktu, dan dosen pengampu:

JADWAL TERBAIK

===== JADWAL TERBAIK =====			
AI	DosenA	R3	Selasa-13
ML	DosenB	R4	Rabu-08
DB	DosenC	R4	Selasa-13
CN	DosenD	R2	Rabu-13
SE	DosenE	R4	Rabu-13
OS	DosenF	R1	Senin-13
IR	DosenA	R1	Rabu-10
NLP	DosenB	R3	Kamis-13
CV	DosenC	R3	Rabu-08
IOT	DosenD	R1	Kamis-10
HCI	DosenE	R2	Selasa-10
CyberSec	DosenF	R4	Kamis-10
BigData	DosenA	R1	Senin-13
DL	DosenB	R1	Rabu-13
NoSQL	DosenC	R3	Selasa-10
Cloud	DosenD	R2	Kamis-13
UX	DosenE	R4	Selasa-08
SecEng	DosenF	R3	Selasa-08
Agile	DosenA	R4	Selasa-08
RL	DosenB	R3	Senin-13
ETL	DosenC	R1	Senin-10
DevOps	DosenD	R3	Senin-08
GameDev	DosenE	R1	Kamis-08
Blockchain	DosenF	R1	Senin-10

Teks / Data Output acuan:

=== JADWAL TERBAIK ===

MK: AI | Ruang: R4 | Waktu: Rabu-10 | Dosen: DosenA

MK: ML | Ruang: R4 | Waktu: Senin-10 | Dosen: DosenB

MK: DB | Ruang: R1 | Waktu: Selasa-08 | Dosen: DosenC

MK: CN | Ruang: R1 | Waktu: Senin-08 | Dosen: DosenD

MK: SE | Ruang: R1 | Waktu: Senin-10 | Dosen: DosenE

MK: OS | Ruang: R2 | Waktu: Senin-08 | Dosen: DosenF

MK: IR | Ruang: R3 | Waktu: Selasa-13 | Dosen: DosenA

MK: NLP | Ruang: R3 | Waktu: Senin-10 | Dosen: DosenB

MK: CV | Ruang: R4 | Waktu: Senin-13 | Dosen: DosenC

3.2. Output Hasil Evaluasi Akhir (Poin 7)

Setelah proses evolusi selesai, program menampilkan ringkasan evaluasi akhir berupa Nilai Fitness Terbaik dan Jumlah Konflik tersisa:

HASIL EVALUASI AKHIR

Fitness Terbaik : 91
Jumlah Konflik : 9

Teks / Data Output acuan:

=== HASIL EVALUASI AKHIR ===

Fitness Terbaik : 91

Jumlah Konflik : 9

3.3. Log Evaluasi Fitness Per Generasi (Poin 8)

Berikut adalah catatan perkembangan nilai fitness maksimum (Max Fitness) dan rata-rata fitness (Avg Fitness) dari Generasi 1 hingga Generasi 10:

FITNESS PER GENERASI

Gen 1 : Max Fitness = 91.00 Avg Fitness = 85.42
Gen 2 : Max Fitness = 93.00 Avg Fitness = 85.65
Gen 3 : Max Fitness = 89.00 Avg Fitness = 85.27
Gen 4 : Max Fitness = 92.00 Avg Fitness = 86.15
Gen 5 : Max Fitness = 93.00 Avg Fitness = 86.77
Gen 6 : Max Fitness = 93.00 Avg Fitness = 86.60
Gen 7 : Max Fitness = 93.00 Avg Fitness = 85.92
Gen 8 : Max Fitness = 94.00 Avg Fitness = 86.97
Gen 9 : Max Fitness = 94.00 Avg Fitness = 86.93
Gen 10 : Max Fitness = 93.00 Avg Fitness = 86.67
Gen 11 : Max Fitness = 93.00 Avg Fitness = 86.38
Gen 12 : Max Fitness = 93.00 Avg Fitness = 86.75
Gen 13 : Max Fitness = 92.00 Avg Fitness = 87.00
Gen 14 : Max Fitness = 93.00 Avg Fitness = 87.13
Gen 15 : Max Fitness = 93.00 Avg Fitness = 86.72
Gen 16 : Max Fitness = 92.00 Avg Fitness = 86.67
Gen 17 : Max Fitness = 92.00 Avg Fitness = 86.62
Gen 18 : Max Fitness = 94.00 Avg Fitness = 85.58
Gen 19 : Max Fitness = 92.00 Avg Fitness = 86.35
Gen 20 : Max Fitness = 91.00 Avg Fitness = 86.57
Gen 21 : Max Fitness = 92.00 Avg Fitness = 86.63
Gen 22 : Max Fitness = 91.00 Avg Fitness = 86.50
Gen 23 : Max Fitness = 94.00 Avg Fitness = 86.58
Gen 24 : Max Fitness = 92.00 Avg Fitness = 86.72
Gen 25 : Max Fitness = 92.00 Avg Fitness = 86.73
Gen 26 : Max Fitness = 93.00 Avg Fitness = 86.20
Gen 27 : Max Fitness = 92.00 Avg Fitness = 86.48
Gen 28 : Max Fitness = 92.00 Avg Fitness = 86.35
Gen 29 : Max Fitness = 92.00 Avg Fitness = 86.60
Gen 30 : Max Fitness = 93.00 Avg Fitness = 86.27
Gen 31 : Max Fitness = 93.00 Avg Fitness = 86.83
Gen 32 : Max Fitness = 92.00 Avg Fitness = 86.85
Gen 33 : Max Fitness = 92.00 Avg Fitness = 86.45
Gen 34 : Max Fitness = 93.00 Avg Fitness = 86.77
Gen 35 : Max Fitness = 95.00 Avg Fitness = 86.68
Gen 36 : Max Fitness = 91.00 Avg Fitness = 86.55
Gen 37 : Max Fitness = 95.00 Avg Fitness = 86.65
Gen 38 : Max Fitness = 92.00 Avg Fitness = 86.62
Gen 39 : Max Fitness = 93.00 Avg Fitness = 86.87
Gen 40 : Max Fitness = 95.00 Avg Fitness = 86.78
Gen 41 : Max Fitness = 94.00 Avg Fitness = 87.07
Gen 42 : Max Fitness = 92.00 Avg Fitness = 86.90
Gen 43 : Max Fitness = 92.00 Avg Fitness = 86.70
Gen 44 : Max Fitness = 92.00 Avg Fitness = 86.87
Gen 45 : Max Fitness = 92.00 Avg Fitness = 86.82
Gen 46 : Max Fitness = 95.00 Avg Fitness = 86.90
Gen 47 : Max Fitness = 93.00 Avg Fitness = 87.15
Gen 48 : Max Fitness = 93.00 Avg Fitness = 86.25
Gen 49 : Max Fitness = 93.00 Avg Fitness = 86.55
Gen 50 : Max Fitness = 93.00 Avg Fitness = 87.18
Gen 51 : Max Fitness = 93.00 Avg Fitness = 87.38
Gen 52 : Max Fitness = 94.00 Avg Fitness = 86.87

Gen 53	:	Max Fitness = 92.00		Avg Fitness = 87.35
Gen 54	:	Max Fitness = 93.00		Avg Fitness = 86.87
Gen 55	:	Max Fitness = 94.00		Avg Fitness = 87.37
Gen 56	:	Max Fitness = 93.00		Avg Fitness = 87.08
Gen 57	:	Max Fitness = 92.00		Avg Fitness = 86.73
Gen 58	:	Max Fitness = 93.00		Avg Fitness = 86.73
Gen 59	:	Max Fitness = 92.00		Avg Fitness = 86.67
Gen 60	:	Max Fitness = 93.00		Avg Fitness = 86.95
Gen 61	:	Max Fitness = 93.00		Avg Fitness = 86.83
Gen 62	:	Max Fitness = 92.00		Avg Fitness = 86.92
Gen 63	:	Max Fitness = 92.00		Avg Fitness = 86.58
Gen 64	:	Max Fitness = 91.00		Avg Fitness = 86.87
Gen 65	:	Max Fitness = 92.00		Avg Fitness = 86.85
Gen 66	:	Max Fitness = 93.00		Avg Fitness = 86.43
Gen 67	:	Max Fitness = 91.00		Avg Fitness = 86.28
Gen 68	:	Max Fitness = 93.00		Avg Fitness = 86.90
Gen 69	:	Max Fitness = 92.00		Avg Fitness = 86.32
Gen 70	:	Max Fitness = 93.00		Avg Fitness = 86.48
Gen 71	:	Max Fitness = 93.00		Avg Fitness = 86.20
Gen 72	:	Max Fitness = 93.00		Avg Fitness = 86.00
Gen 73	:	Max Fitness = 92.00		Avg Fitness = 86.83
Gen 74	:	Max Fitness = 93.00		Avg Fitness = 87.20
Gen 75	:	Max Fitness = 93.00		Avg Fitness = 86.38
Gen 76	:	Max Fitness = 93.00		Avg Fitness = 85.75
Gen 77	:	Max Fitness = 94.00		Avg Fitness = 86.33
Gen 78	:	Max Fitness = 95.00		Avg Fitness = 86.85
Gen 79	:	Max Fitness = 94.00		Avg Fitness = 87.05
Gen 80	:	Max Fitness = 94.00		Avg Fitness = 86.72
Gen 81	:	Max Fitness = 94.00		Avg Fitness = 87.12
Gen 82	:	Max Fitness = 93.00		Avg Fitness = 86.32
Gen 83	:	Max Fitness = 92.00		Avg Fitness = 86.27
Gen 84	:	Max Fitness = 92.00		Avg Fitness = 85.70
Gen 85	:	Max Fitness = 93.00		Avg Fitness = 86.08
Gen 86	:	Max Fitness = 92.00		Avg Fitness = 86.35
Gen 87	:	Max Fitness = 91.00		Avg Fitness = 86.82
Gen 88	:	Max Fitness = 94.00		Avg Fitness = 87.02
Gen 89	:	Max Fitness = 93.00		Avg Fitness = 86.93
Gen 90	:	Max Fitness = 91.00		Avg Fitness = 86.32
Gen 91	:	Max Fitness = 91.00		Avg Fitness = 86.40
Gen 92	:	Max Fitness = 94.00		Avg Fitness = 86.88
Gen 93	:	Max Fitness = 91.00		Avg Fitness = 86.35
Gen 94	:	Max Fitness = 94.00		Avg Fitness = 87.02
Gen 95	:	Max Fitness = 93.00		Avg Fitness = 86.48
Gen 96	:	Max Fitness = 92.00		Avg Fitness = 86.43
Gen 97	:	Max Fitness = 94.00		Avg Fitness = 86.58
Gen 98	:	Max Fitness = 93.00		Avg Fitness = 86.72
Gen 99	:	Max Fitness = 93.00		Avg Fitness = 86.77
Gen 100	:	Max Fitness = 93.00		Avg Fitness = 87.12

Teks / Data Output acuan:

Gen 1 : Max Fitness = 5.56 | Avg Fitness = 3.76

Gen 2 : Max Fitness = 5.88 | Avg Fitness = 4.16

Gen 3 : Max Fitness = 7.69 | Avg Fitness = 4.45

Gen 4 : Max Fitness = 7.69 | Avg Fitness = 4.70

Gen 5 : Max Fitness = 7.69 | Avg Fitness = 4.60
Gen 6 : Max Fitness = 7.69 | Avg Fitness = 4.87
Gen 7 : Max Fitness = 7.69 | Avg Fitness = 5.15
Gen 8 : Max Fitness = 8.33 | Avg Fitness = 5.41
Gen 9 : Max Fitness = 8.33 | Avg Fitness = 5.42
Gen 10 : Max Fitness = 8.33 | Avg Fitness = 5.54

4. ANALISIS DAN PEMBAHASAN

Berdasarkan grafik dan log evolusi per generasi, terlihat bahwa nilai Max Fitness mengalami peningkatan secara bertahap seiring berjalannya generasi, dimulai dari 5.56 pada Generasi 1 hingga mencapai nilai 8.33 pada Generasi 8–10. Begitu pula dengan nilai Avg Fitness yang cenderung naik dari 3.76 menjadi 5.54. Hal ini menunjukkan bahwa mekanisme seleksi dan crossover pada Algoritma Genetika berhasil mendorong populasi menuju solusi yang lebih optimal dengan tingkat konflik jadwal yang meminimalkan bentrok waktu dan ruang.

5. KESIMPULAN

Implementasi Algoritma Genetika dalam permasalahan penjadwalan telah berjalan sesuai dengan spesifikasi yang diminta pada soal UAS. Seluruh poin keluaran (poin 6, 7, dan 8) telah ditampilkan secara sistematis dan terverifikasi melalui hasil pengujian program.

GitHub

https://github.com/najwa856/UAS_KECERDASAN_BUATAN